

1	2	3
Prefer Initial Condition	Prefer Alternative A	Prefer Alternative B

Please press a number from 1 to 3 that best reflects your choice if faced with these options.

a relatively high cost. In essence, subjects can choose to remain in a relatively severe condition and not pay any additional costs for treatments outside the government health plan or insurance plan, or they can choose to pay for additional treatments to improve their health to their current health state.

both of the hypothetical alternatives. In such cases, at least one of the options would be preferred, or dominant, in all of the attributes, and thus choices would reveal less information about trade-off relations. This situation does not invalidate the design, but introduces some degree of statistical inefficiency.

The health history questions collect information on the subject's personal health history, the health history of family members and socio-demographic information about the subject and the subject's household. The information gathered is used in

the analysis to determine whether health history and other demographic variables influence WTP for health. This information also was used to characterize the subject's current health condition in Alternative B of the choice sets. The instrument was administered to 399 randomly recruited subjects in the Toronto area between March and July 1997.⁹

ANALYSIS OF GRADED-PAIR DATA

Graded-pair responses are ordinal ratings of utility differences between attribute level pairs. Estimation strategy should thus account for the discrete, ordinal nature of the response variable. In this section, we describe a general model for estimating subject utility functions from such data and a procedure for calculating marginal WTP using the estimated utility functions.^r We assume that individual indirect utility can be expressed as a function of commodity attributes and personal characteristics:

$$U_{st}^i = V^i(X_{st}, Z^i, P_{st}; \beta^i, \delta^i) + e_{st}^i \quad (1)$$

where U_{st}^i is individual i 's utility for commodity profile st , where $s = L$ or R , denoting the left side and right side profiles for pair t , $t = 1, \dots, 8$; $V^i(\cdot)$ is the non-stochastic part of the utility function; X_{st} is a vector of attribute levels in profile st ; Z^i is a vector of personal characteristics; P_{st} is the cost of the commodity profile; β^i is a vector of attribute parameters; δ^i is the marginal utility of money; and e_{st}^i is a disturbance term.

The utility difference for profile pair t , dU_t^i , is simply

$$dU_t^i = V_{Rt}^i - V_{Lt}^i + \varepsilon_t^i \quad (2)$$

where V_{Rt}^i and V_{Lt}^i are the indirect utilities associated with the right side and left side profiles, respectively, and $\varepsilon_t^i = e_{Rt}^i - e_{Lt}^i$ is the associated disturbance term. The disturbance term captures the effects of unobserved factors, including possible inherent ambiguity of subject preferences and cognitive errors.

The difference in indirect utility for commodity pair t , dU_t^i , often is specified as a simple linear function of attributes

$$\begin{aligned} dU_t^i &= V_{Rt}^i - V_{Lt}^i + \varepsilon_t^i \\ &= \left[\sum_h \beta_h \cdot X_{hRt} + \delta \cdot P_{Rt} \right] \\ &\quad - \left[\sum_h \beta_h \cdot X_{hLt} + \delta \cdot P_{Lt} \right] + \varepsilon_t^i \end{aligned} \quad (3)$$

where h indexes attributes. This specification assumes that attributes are neither substitutes nor complements for each other, so a change in the level of one attribute does not affect the marginal utility of any other attribute.^s

The effects of personal characteristics on utility differences do not appear in Equation (3) because subject personal characteristics do not vary between the left and right sides of the screen. Controlling for such variables requires interacting them with commodity attributes or prices that do vary between profiles. For example, we can estimate the marginal utility of money as a function of individual characteristics Z^i and employ a functional form that allows for diminishing marginal utility of money.

$$\begin{aligned} dU_t^i &= V_{Rt}^i - V_{Lt}^i + \varepsilon_t^i \\ &= [X_{Rt} \cdot \beta + (Z^i \cdot \gamma) \cdot \sqrt{P_{Rt}}] \\ &\quad - [X_{Lt} \cdot \beta + (Z^i \cdot \gamma) \cdot \sqrt{P_{Lt}}] + \varepsilon_t^i \end{aligned} \quad (4)$$

where γ is a vector of marginal utility of money parameters.^t

Similarly, the β coefficients on attributes could also be allowed to vary across subjects by making them functions of individual characteristics. It is also possible to employ more general non-linear specifications of continuous attribute variables or interactions among attribute variables.

We do not observe dU_t^i directly. Instead, we observe C_t^i , which is a discrete rating category related to the unobserved dU_t^i of interest. The appropriate approach, therefore, is ordered logit or probit, which incorporates both the discreteness and the natural ordering of the data. This study uses ordered probit, which assumes the error term is normally distributed. To estimate ordered probit models, the data are sorted so that the preferred profile is on the right, making $dV_t^i = V_{Rt}^i - V_{Lt}^i \geq 0$.^u We construct the rating categories by recoding responses accordingly, so that 0 indicates indifference and 3 indicates maximum difference.^v Because probit assumes the Equation (3) error term ε_t^i is distributed $N(0, \sigma^2)$, the probability of observing response C_t^i is

$$\text{Prob}(C_t^i = k) = \Phi[\mu^i \cdot (\alpha_k - dV_t^i)] - \Phi[\mu^i \cdot (\alpha_{k-1} - dV_t^i)], \quad k = 0, 1, \dots, 3, \quad (5)$$

where Φ is the cumulative standard normal distribution function, the α_k are threshold constants, and scale parameter μ^i is the inverse of the standard deviation.^w

Because the health valuation survey collects eight responses from each subject, it is appropriate to estimate a panel model, which accounts for correlated errors in each subject's series of ratings. A random effects model incorporates an individual-specific error term, so that

$$\varepsilon_t^i = \eta_t^i + \lambda^i, \quad (6)$$

where η_t^i is a common error term distributed by $N(0, \sigma_\eta^2)$, λ^i is an individual-specific error term distributed by $N(0, \sigma_\lambda^2)$ and $\sigma_\varepsilon^2 = \sigma_\eta^2 + \sigma_\lambda^2$. The correlation between successive disturbances for the same subject is ρ .

Most ordered-category data contain no information on how variance or scale might vary across subjects, and thus it is usually normalized uniformly to one. However, graded-pair data include both cross-section and time-series information, thus it is possible to obtain scale estimates. We can account for non-uniform scale by letting ε_t^i vary across subjects so that $\varepsilon_t^i \sim N(0, 1/\mu_i^2)$.^x Alternatively, we can make μ_i a systematically varying parameter that is a function of personal characteristics Y^i , such as income and education, so that $\varepsilon_t^i \sim N[0, 1/\mu(Y^i)^2]$.

Integrating out the random effect, Equation (5) now becomes

$$\begin{aligned} \text{Prob}(C^i = k_1^i, k_2^i, \dots, k_8^i) \\ = \int_{-\infty}^{\infty} \phi(\lambda^i) \cdot \prod_{t=1}^8 \Phi[\mu^i \cdot (\alpha_{k_t} - dV_t^i)] \\ - \Phi[\mu^i \cdot (\alpha_{k_t-1} - dV_t^i)] d\lambda^i, \end{aligned} \quad (7)$$

where $\phi(\lambda^i)$ is the normal probability density function for λ^i .^y

Estimating individual scale parameters provides a means of quantifying the coherence of individual rating patterns and of controlling for different levels of variance in subjects' error terms. Subjects who refuse to solve or have difficulty solving the utility difference problem and who enter random or repetitive ratings will have unusually noisy ratings and thus larger estimated variance. Scale estimates thus provide a means of testing whether groups of subjects are experiencing problems with

the survey design. The consistency of a subject's rating pattern can vary by the degree of attentiveness, quantitative orientation, age, educational background, susceptibility to fatigue, and other factors. In addition, we can identify individual subjects who may have failed to perform the rating task properly. Finally, individual scale estimates help account for differences in how subjects interpret and use the 0-to-3 rating categories. Some subjects may be reluctant to use the entire range of ratings, which is a common problem in graded-pair elicitations [18,19].

ANALYSIS OF DISCRETE-CHOICE DATA

The SP survey also included a series of choice judgements with three alternatives in each choice set. The linear specification of utility analogous to Equation (4) for the three alternatives is

$$\begin{aligned} U_{jt}^i &= V_{jt}^i + \varepsilon_{jt}^i = X_{jt} \cdot \beta + (Z^i \cdot \gamma) \sqrt{P_{jt}} + \varepsilon_{jt}^i, \quad j = 1, 2; \\ U_{jt}^i &= V_0^i + \delta_t^i + \varepsilon_{0t}^i = H^i \cdot \omega + (Z^i \cdot \gamma) \sqrt{P_{jt}} + \varepsilon_{jt}^i, \\ &j = 3, \end{aligned} \quad (8)$$

where U_{jt}^i , $j = 1, 2$, is the utility of each of the two alternative health profiles, and U_{jt}^i , $j = 3$, is the utility of the current health alternative. V_0^i is the subject's status quo health state utility, H^i is a vector of health history variables and ω are associated parameters. Other variables were defined previously.

Assuming ε follows a type 1 extreme value error structure, the probability that alternative j will be selected from choice set t is the standard conditional-logit expression

$$\text{Prob}(C_t^i = j) = \frac{\exp(\mu^i \cdot V_{jt}^i)}{\sum_{k=1}^3 \exp(\mu^i \cdot V_{kt}^i)}, \quad (9)$$

where C_t^i is the selected alternative in each of eight choice sets, μ^i is the scale parameter and V_j^i is the determinate part of the utility of alternative j .^z Note that individual characteristics again fall out of this expression unless interacted with health attributes. The scale parameter μ^i also is generally not identifiable in such models and is normalized at 1. However, because we have

multiple observations for each subject, we can estimate $\mu^i = \mu(Z^i)$. Like the ordered-probit analysis described above, the conditional logit model specified by Equations (8) and (9) is estimated using maximum likelihood.^{aa}

Conditional logit models are known to be subject to violations of the irrelevance of independent alternatives assumption (IIA). This condition requires that the ratio of probabilities for any two alternatives be independent of the attribute levels in the third alternative. Tests of the choice data indicate that conditional logit estimates do not satisfy this requirement. Under these conditions, parameter estimates are biased. Second, the conditional logit models assume that differences in WTP across subjects arise only from differences in the marginal utility of money. Finally, conditional logit does not account for correlations within each subject's series of choices. Revelt and Train [20] recently have proposed using random parameter logit (RPL) for SP data similar to ours. RPL is not subject to the IIA assumption, accommodates correlations among panel observations and accounts for uncontrolled heterogeneity in tastes across subjects.

Modifying Equation (8) to introduce subject-specific stochastic components for each β

$$U_{jt}^i = V_{jt}^i + \varepsilon_{jt}^i \equiv X_{jt} \cdot (\beta + \eta^i) + (Z^i \cdot \gamma) \sqrt{P_{jt}} + \varepsilon_{jt}^i. \quad (10)$$

Equation (9) now becomes

$$\text{Prob}[C^i = (C_{j1}^i, C_{j2}^i, \dots, C_{j8}^i)]$$

$$= \prod_{t=1}^8 \left[\frac{\exp[\mu^i \cdot V_{jt}^i(\beta^*)]}{\sum_{k=1}^3 \exp[\mu^i \cdot V_{kt}^i(\beta^*)]} \right] \quad (11)$$

where now $\beta^* = (\beta + \eta^i)$. In contrast to conditional logit, the stochastic part of utility now may be correlated among alternatives and across the sequence of choices via the common influence of η^i . McFadden and Train [21] show that any random utility model can be approximated by some RPL specification.

CALCULATING WILLINGNESS TO PAY

Estimating the parameters of the utility function enables us to quantify the value of changes in commodity attributes. The units of the estimated

utility index essentially are arbitrary. However, this arbitrary measure can be converted into a dollar metric using the estimated marginal utility of money. Let X_j^0 indicate the status quo vector of attribute levels. In our case, this corresponds to the subject's current health state. The cost of X_j^0 is P_j^0 , which we take to be zero. X_j^* indicates a changed vector of attribute levels corresponding to a given combination of symptom and activity level. The WTP for a given change in commodity attributes ($X_j^* - X_j^0$) is the amount of money ($P_j^* - P_j^0$) that would leave subject i indifferent between the payment and the change in attribute levels, so that P_j^* satisfies

$$\begin{aligned} V^i[X_j^*, Z^i, P_j^*; \beta^i, \delta^i(P, Z^i)] \\ = V^i[X_j^0, Z^i, P_j^0; \beta^i, \delta^i(P, Z^i)]. \end{aligned} \quad (12)$$

In the linear specification of Equations (4) and (10), the β coefficient for each attribute represents its constant marginal utility. The negative of the coefficient of the price attribute is interpreted as the marginal utility of money, or the utility derived from more dollars. Thus,

$$\begin{aligned} \text{WTP}^i(X_j^* - X_j^0) &= \frac{\sum_h (X_{hj}^* - X_{hj}^0) \frac{\partial V^i}{\partial X_{hj}}}{-\frac{\partial V^i}{\partial P}} \\ &= \frac{\sum_h (X_{hj}^* - X_{hj}^0) \beta_h^i}{\frac{-(Z^i \cdot \gamma)}{(2 \cdot \sqrt{P})}}, \end{aligned} \quad (13)$$

where h indexes attributes (McFadden [22]).

Allowing the marginal utility of money to vary with personal characteristics allows for differences in tastes to affect the relative utilities of health and money, even though the β_j parameters are constant across subjects. This specification facilitates transferring WTP estimates to populations with different demographic characteristics than our sample. Thus, because WTP estimates are obtained by dividing the health coefficients by the marginal utility of money, changing values of the personal characteristics included in the utility of money function affects the WTP estimates.

Any payment less than or equal to $\text{WTP}^i(X_j^* - X_j^0)$ leaves individuals at least as well-off as they would be if the change ($X_j^* - X_j^0$) had not occurred. Because we recover the parameters for a complete utility index, WTP can be constructed for any utility difference, assuming that the